Controllability, Observability and Balanced Realisations

Multivariable Control, Homework 6

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English summary of my original Persian report, which is kept as a PDF in this folder.

- **Near pole–zero cancellation.** I compared the step responses of $G(s) = 1.01/(s+2)$ and $G(s) = (s+1.01)/((s+1)(s+2))$. The near cancellation makes the second-order system behave almost like the first-order one.
- **Exercises 4-1 and 4-2** (Khaki-Sedigh, *Multivariable Control*). For the given system I checked with MATLAB that it is controllable, observable, functionally controllable ($\det C(sI-A)^{-1}B \neq 0$) and output controllable (rank 3). I also computed the controllability and observability indices from the matrices Φ_c and Φ_o: both sets are $\{2,1,2\}$, summing to 5.
- **Exercises 4-3 and 4-4** from the same book.
- **MATLAB's Lyapunov solver.** I checked which papers the `lyap`/`dlyap` solvers are based on: Bartels–Stewart (1972), Barraud (1977), Hammarling (1982), Penzl (1998), and Golub–Nash–Van Loan (1979).
- **Hankel singular values.** I derived the relations between the gramians and the Hankel singular values used in balanced truncation (slide 15, session 10).